FLASH DROUGHT PROJECT MEETING #11

Jess Bhardwaj
Mike Hobbins
David Hoffmann
Tess Parker

- Jess: presented some slides.
- Ref ET climatology – redone for short crop to compare to tall crop, very similar
- Reran for BARRA-C2 winds (higher resolution)– ref ET climatology, and comparison to AWO: perhaps better biases along the coastline.
- Might plot the percentile climatology, as well as mean climatology and variability difference.
- Most dominant driver seasonally from contributions – some interesting regional differences.
- BARRA-C2 higher resolution runs still to do, but computationally crashes.
- Mike: seasonal plots of sensitivity multiplied by anomalies across whole period, daily contributions from each driver, by season = seasonally averaged dominating daily driver. Almost a variability but doesn't have covariances between drivers.
- Mike: you have 6 data sets with contributions on a daily basis. We have the space and time of droughts and flash droughts; need to identify events. Also do the double standardisation for extremes of the extremes.
- Mike's 2025 paper – focus was impact on crops, therefore thirstwaves rather than droughts.
- Jess has a skeleton writeup. Method section particularly important to get right (standardisation, double standardisation, thirstwaves).
- Have been discussions with Naomi from the ACS fire team – interested to look at a more tropical event. In Black Summer, the Daintree Forest burned for the first time. Naomi will look for notable fires from that area.
- Important to do composite maps and then event details.
- Jess also keen to do covariance plots for drivers.
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- David – adding increasing CO_2 to PET using Yang et al (2019) formulation.
- Less difference in energy limited regions compared to water limited. Largest difference in the interior of Australia.
- Plan to broadcast over the ACS ensemble of RCMs.
- Mike: could add CO_2 as a driver, as it affects only one factor in the denominator of the PET formula. Particularly relevant for projections space.
- Mike: Flash droughts in a hotter future may be a dramatic future. Look at the climatology under a hotter but more CO_2 future. That's a Nature paper! CO_2 has a mitigating effects on past trends so this would be interesting.
- Mike will do some maths to pull CO_2 out as a driver.

Jess is away from 1 June to 17 July. Mike is away in July. David goes on leave for 3 weeks from 5 August. We'll try to meet on Monday 3 / Tuesday 4 August to catch up.